Computational Biology

Lecture 1: *Introduction*

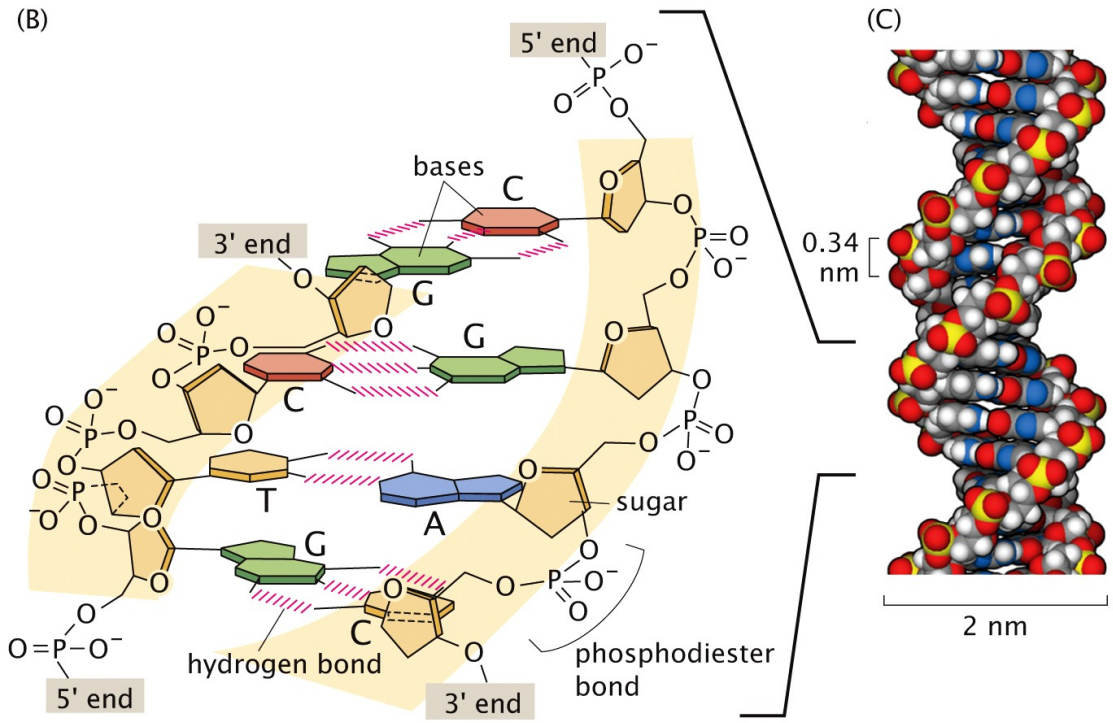

Figure 1.3b Physical Biology of the Cell, 2ed. (© Garland Science 2013)

Evolution, DNA, Transciption, Translation

Outline

1. **Big Data**
2. Beginnings
3. DNA
4. Protein Production

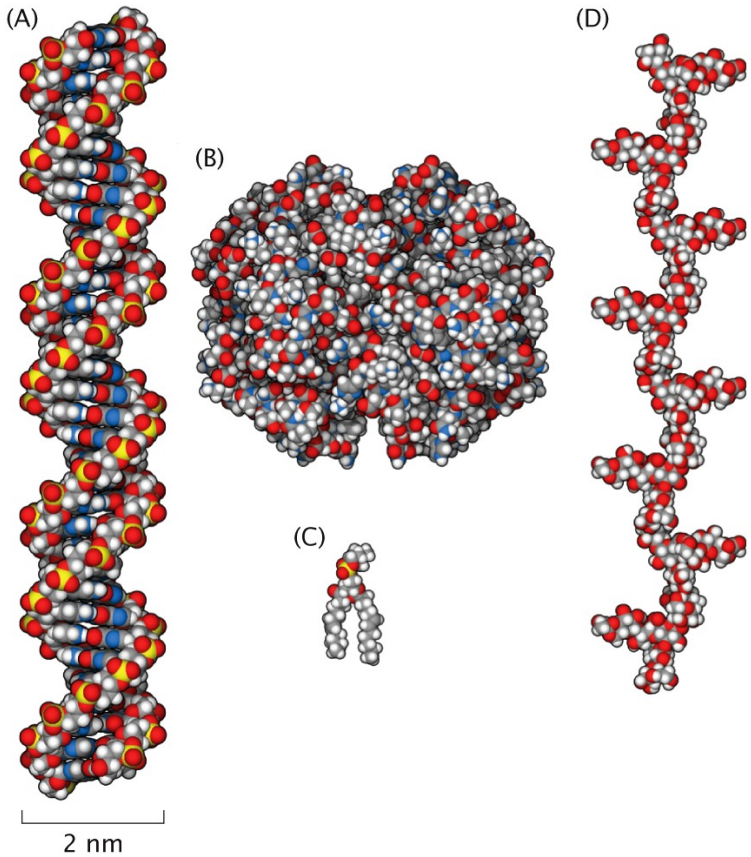

Figure 1.1 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

Biology is a Big Data Problem!

- Biology is big
  - ⋆ Genomes involve 10^5–10^{12} nucleotide bases
  - ⋆ Number of different proteins can be 10^5
  - ⋆ Number of different cell types 10^2
  - ⋆ Number of species 10^7
- Understanding biology involves a lot of information
- This information is stored in big databases

Interactions

- Interactions occur across many levels
  - ⋆ Protein-protein interaction are common place
  - ⋆ Protein-DNA interaction underlie a lot of regulation
  - ⋆ Protein-environment interaction are how cell sense the world
  - ⋆ Species interact in ecosystems
    - ∗ You are an ecosystem
    - ∗ Half the cells in your body aren't yours
- If $P \approx 10^5$ is the number of proteins then the number of possible interacting pairs are $P \approx 10^{10}$

Biology Self-Organise

- Protein complexes form machines
- Life organises into cells
- Cells can organise into higher organisms
- Organisms form populations that act together
- Back to ecosystems

Life is Dynamic

- Cells are chemical machines
- They work through complex auto-regulation
- Cells divide
- Sometimes cell fuse sexually
- Life evolves
- Almost all aspects of life are governed by control mechanisms

Computational Biology

- Making sense of biology is a big data problem
- Tools of computer science are essential to
  - ⋆ store the data
  - ⋆ make inferences about the data
- The tools of abstraction applies to biology as much as to computer science
- Biology is both infinitely complex and yet in some cases amazing simple

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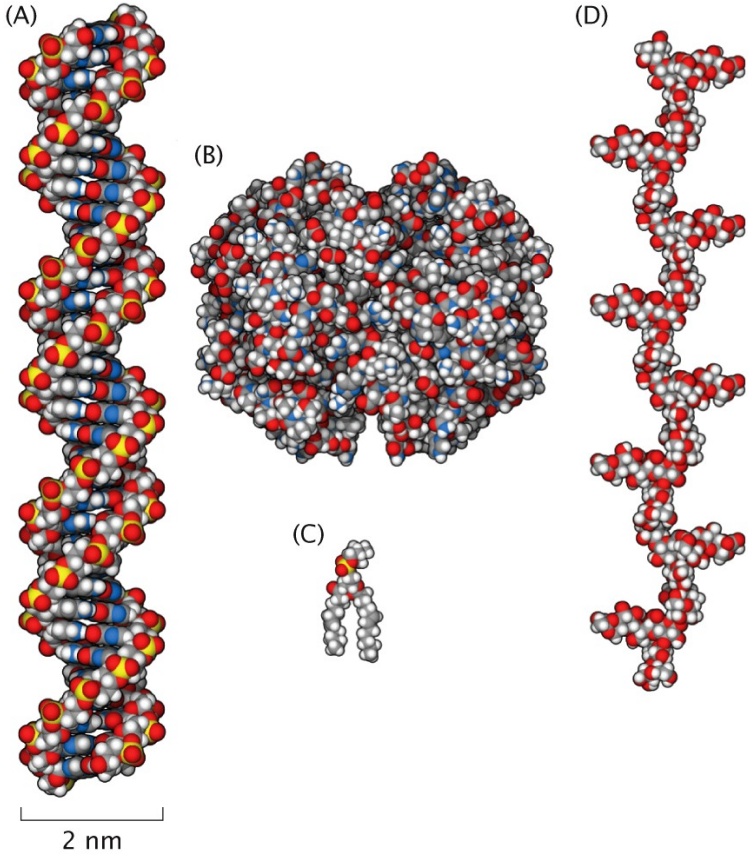
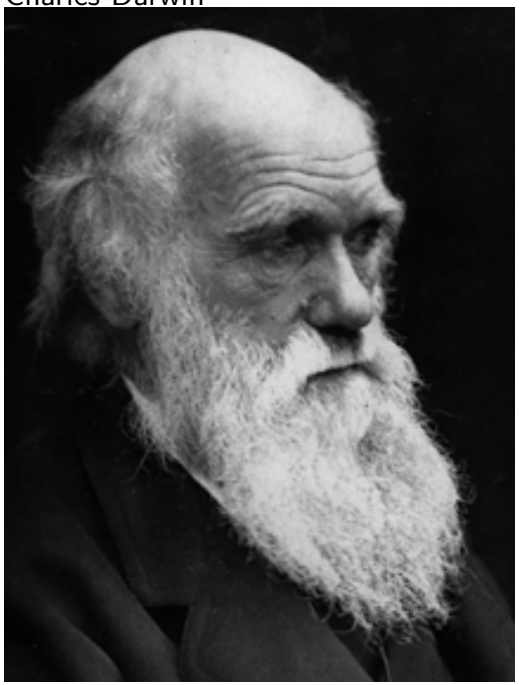


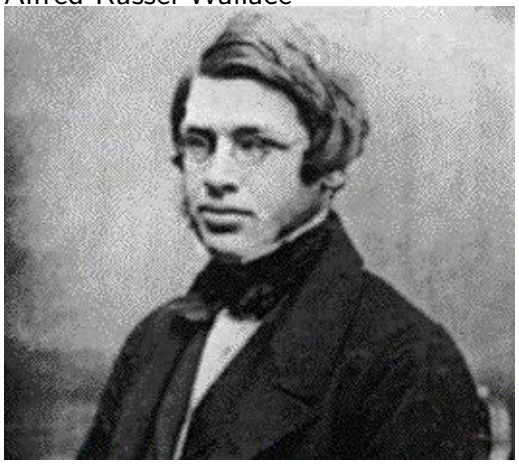
Figure 1.1 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

Evolution

Charles Darwin

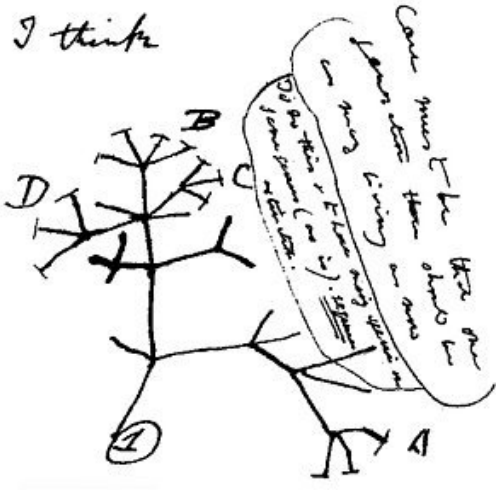


Alfred Russel Wallace



- "Nothing in Biology Makes Sense Except in the Light of Evolution"
  –Theodosius Dobzhansky
- At every level (from protein to species) we can make sense of biology in term of natural selection
- Understanding the details of evolution is an on-going story

The Tree of Life

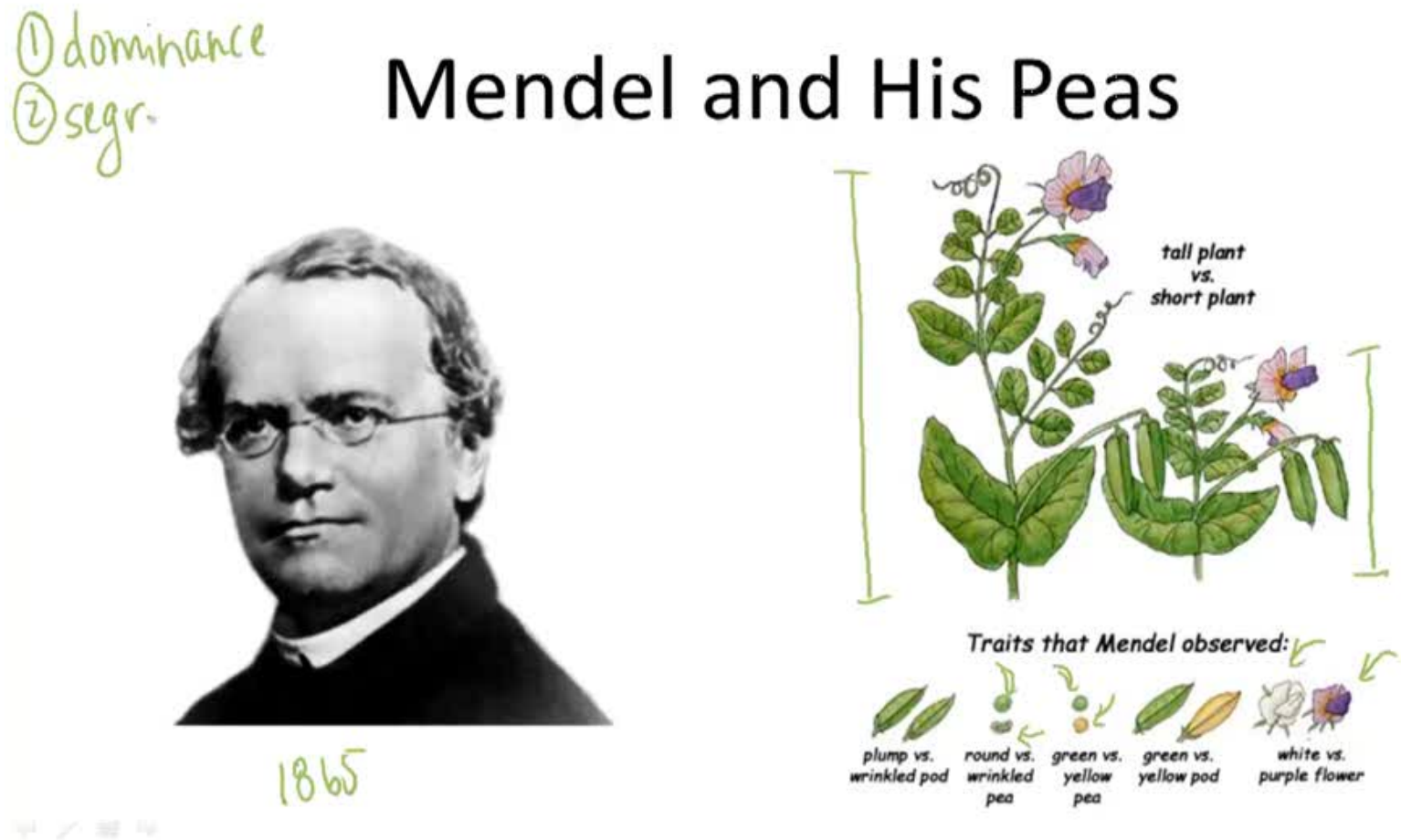


- All observations are consistent with life on earth being related through a tree like structure
- At the molecular level different species share a huge amount in common
- Biologists use this to transfer knowledge between species

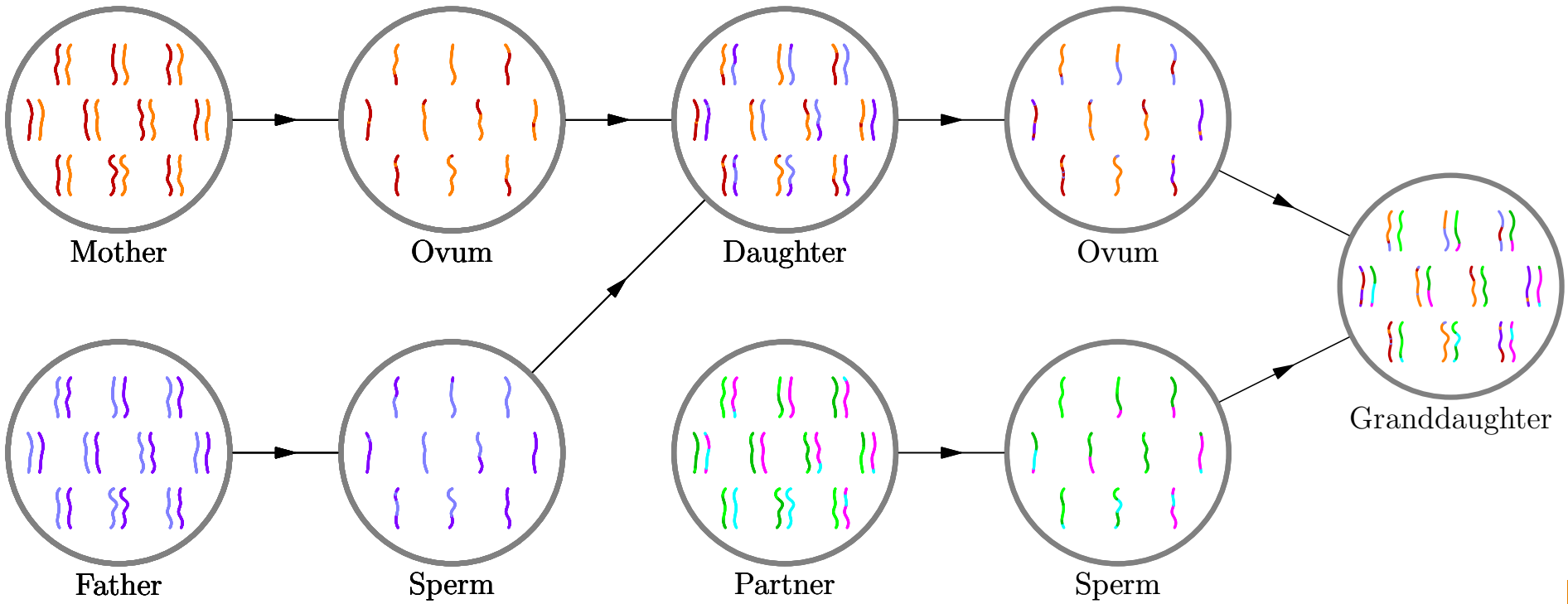
Inheritance

- At the time of Darwin (1859) inheritance was **not** generally understood
- This caused a lot of confusion because blending inheritance would seem to remove the variation needed for natural selection
- Modern inheritance theory was initially proposed by Gregor Mendel around 1860, but was not widely known
- Around 1900 it was finally widely understood that inheritance occurred through the exchange of discrete units (genes)

Gregor Mendel's Experiment



Sexual Reproduction



Mendel's Peas

Inheritance

- Mendel's experiments could be explained if
  - ⋆ traits were carried by discrete units (genes)
  - ⋆ each individual carries two copies of a gene (diploidy)
  - ⋆ we inherit one copy from each parent
  - ⋆ one gene is dominant
- These are not always true
- Very clever explanation and experiments, although the results were rigged
- The mechanism for inheritance remained confused until. . .

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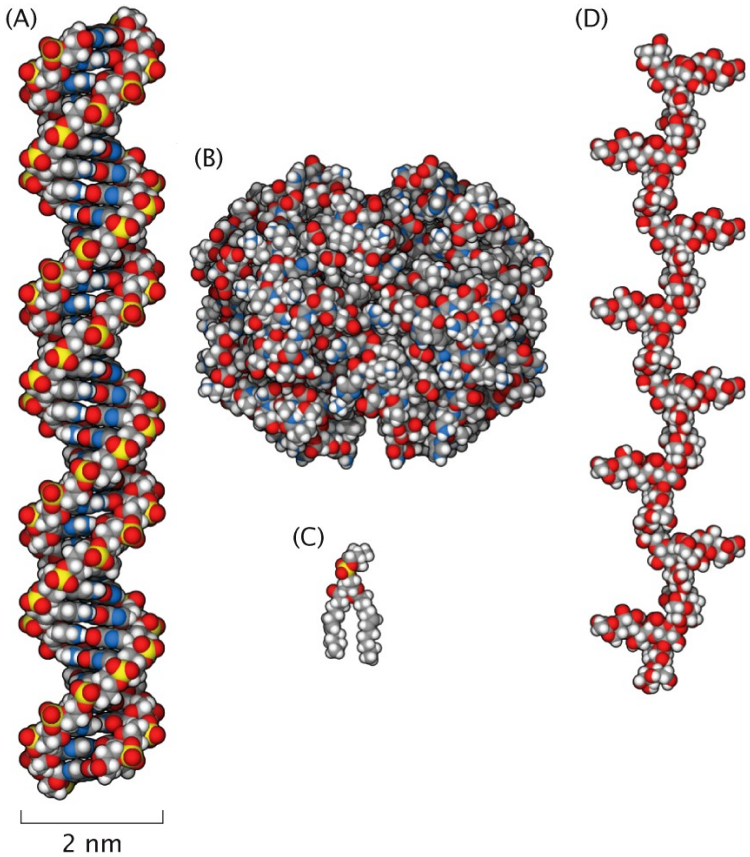


Figure 1.1 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

Crick and Watson

Discovery of the Double Helix

No. 4356 April 25, 1953 NATURE

equipment, and to Dr. G. E. R. Deacon and the captain and officers of R.R.S. *Discovery II* for their part in making the observations.

[1] Young, F. B., Gerrard, H., and Jevons, W., *Phil. Mag.*, **40**, 149 (1920).

[2] Longuet-Higgins, M. S., *Mon. Not. Roy. Astro. Soc., Geophys. Supp.*, **5**, 285 (1949).

[3] Von Arx, W. S., Woods Hole Papers in Phys. Oceanog. Meteor., **11** (3) (1950).

[4] Ekman, V. W., *Arkiv. Mat. Astron. Fysik. (Stockholm)*, **2** (11) (1905).

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

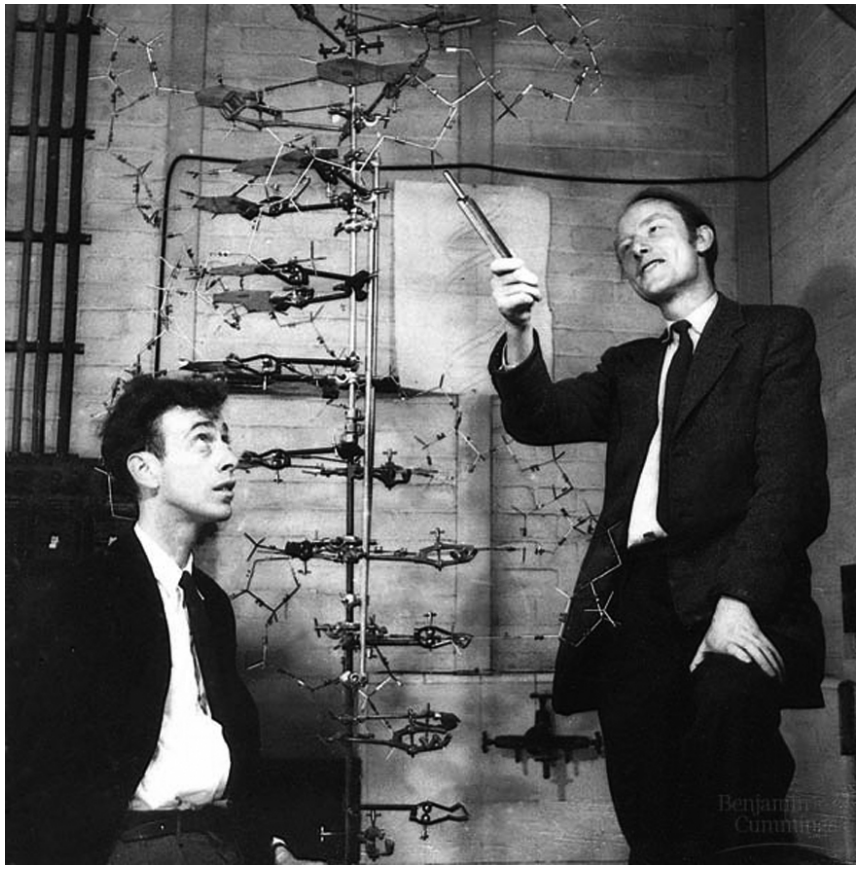
WE wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

A structure for nucleic acid has already been

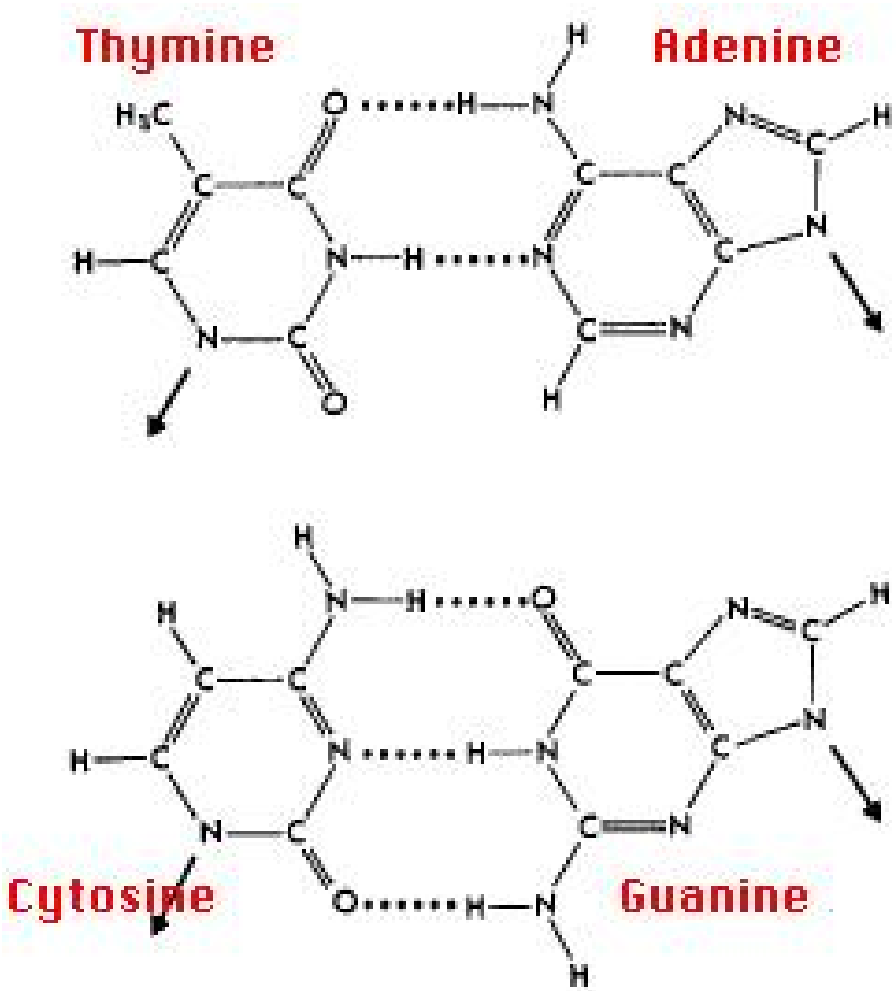
is a residue on each
tion. We have as
adjacent residues i
structure repeats aft
is, after 34 A. The
from the fibre axis i
the outside, cations

The structure is a
is rather high. At
expect the bases to
become more comp

The novel feature
in which the two
purine and pyrimidi
are perpendicular to
together in pairs, a
hydrogen-bonded to
chain, so that the t
z-co-ordinates. One
the other a pyrimi
hydrogen bonds are
1 to pyrimidine p

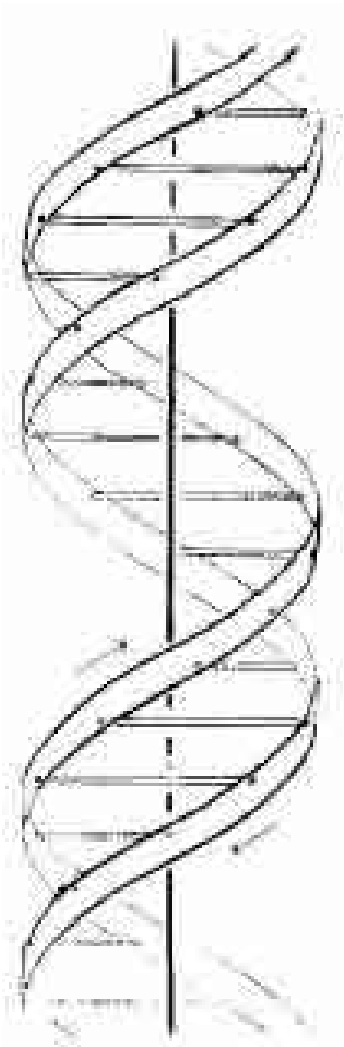


Base Pairs



- Double helix held together by 4 types of bases
- Each base, A, C, G, T form **base pairs** A–T or C–G
- Amazing digital code

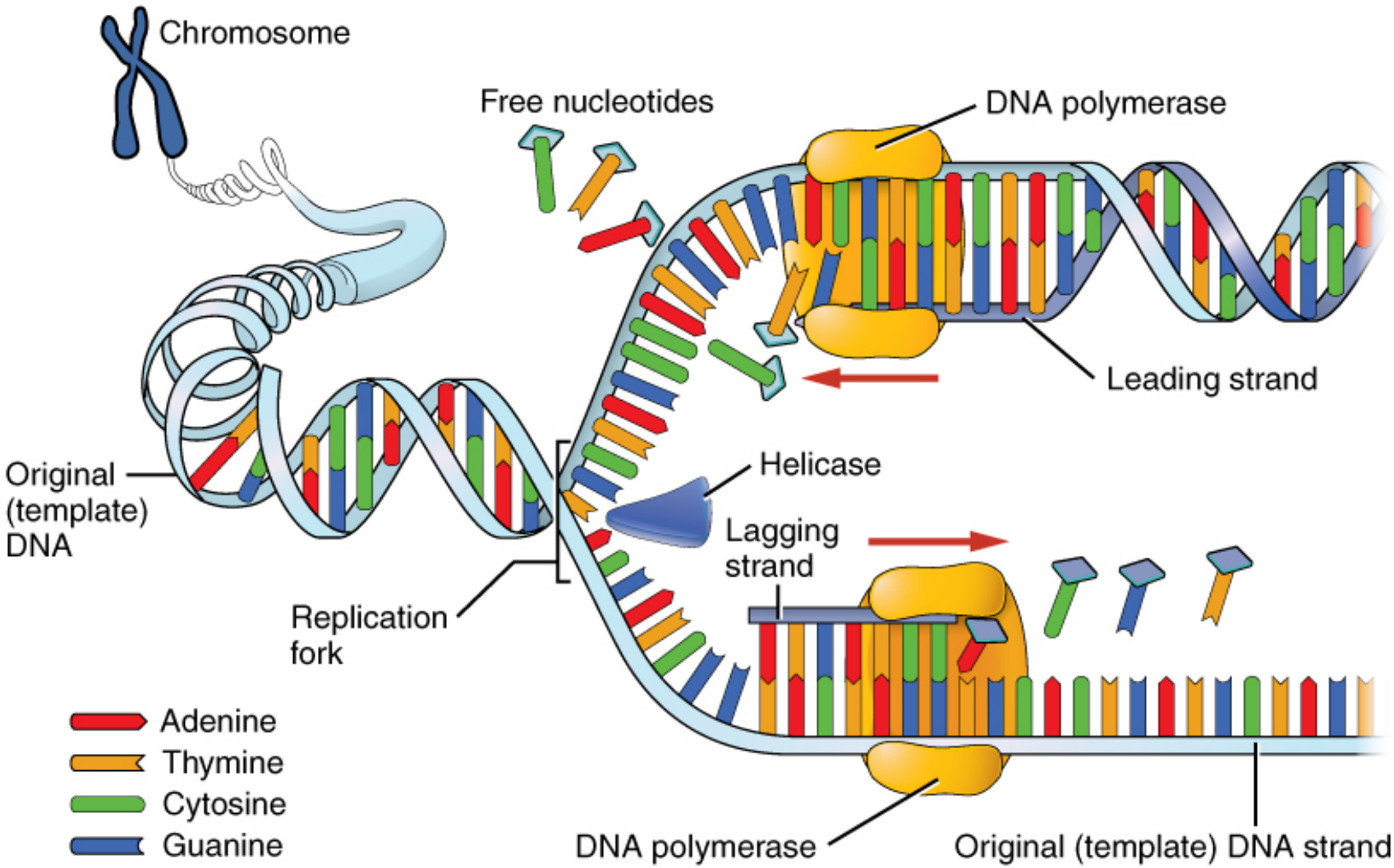
The Genetic Code



"It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material."

– Crick and Watson (1953)

DNA Replication



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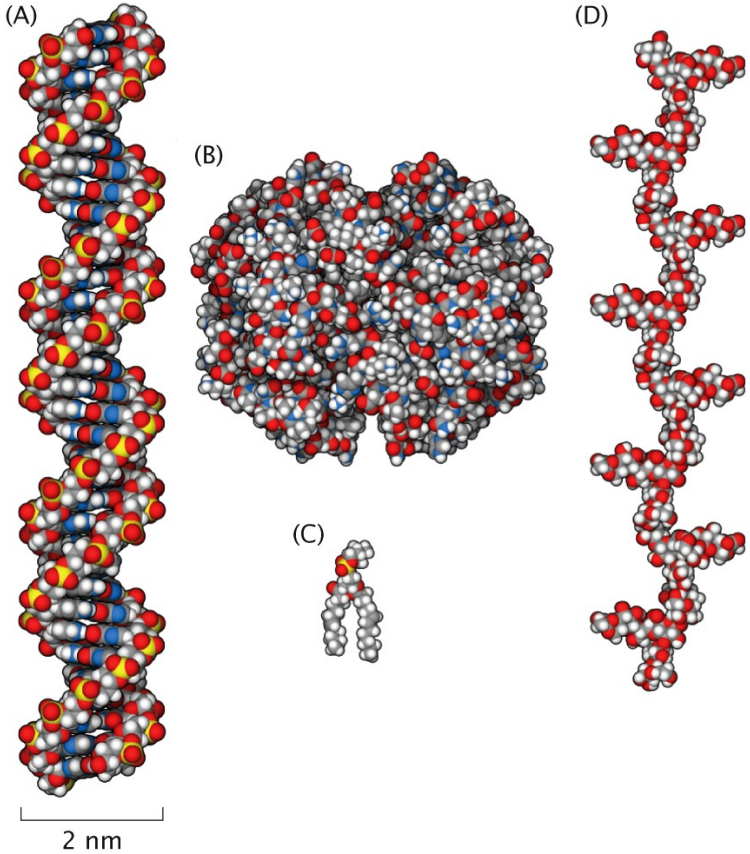
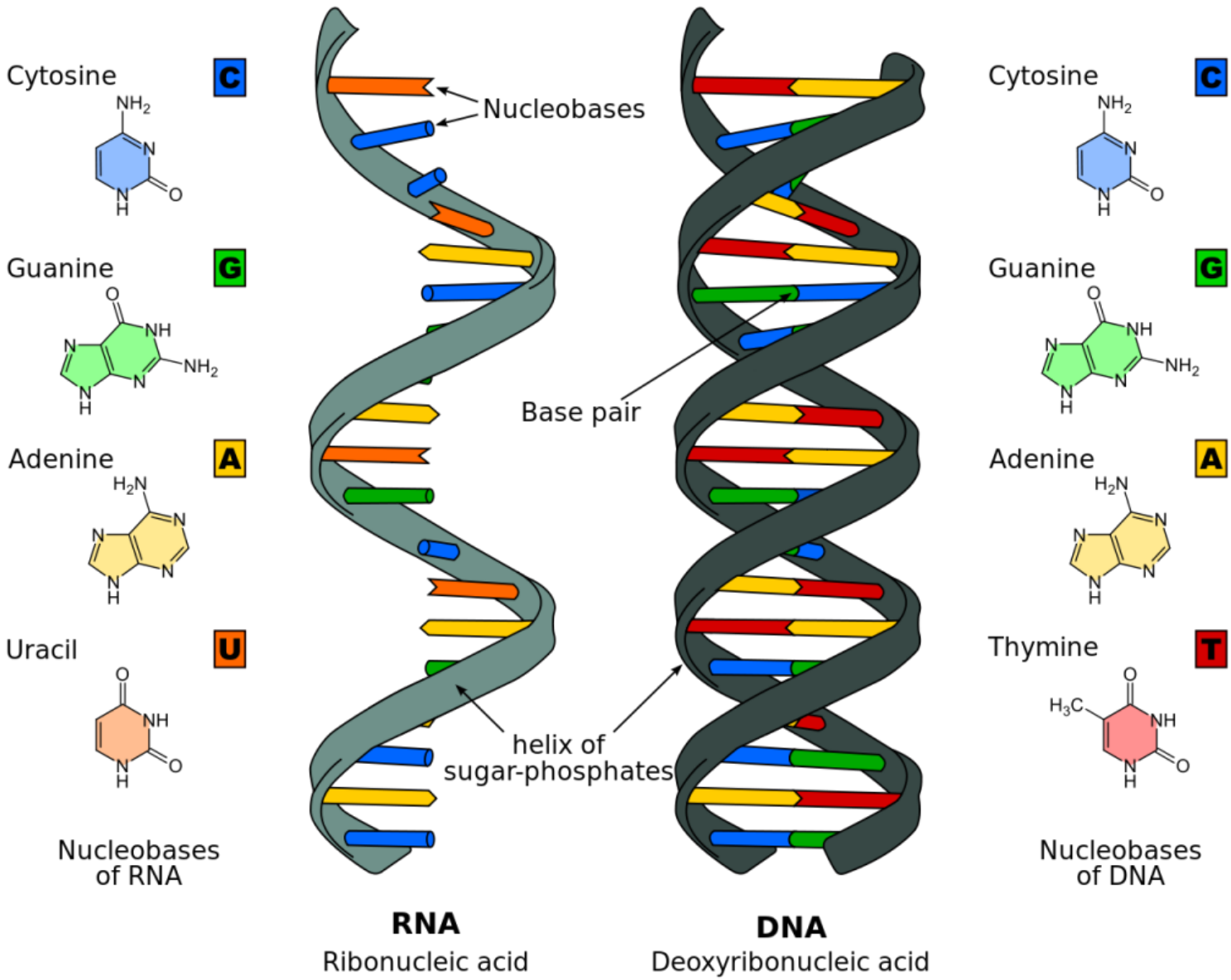


Figure 1.1 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

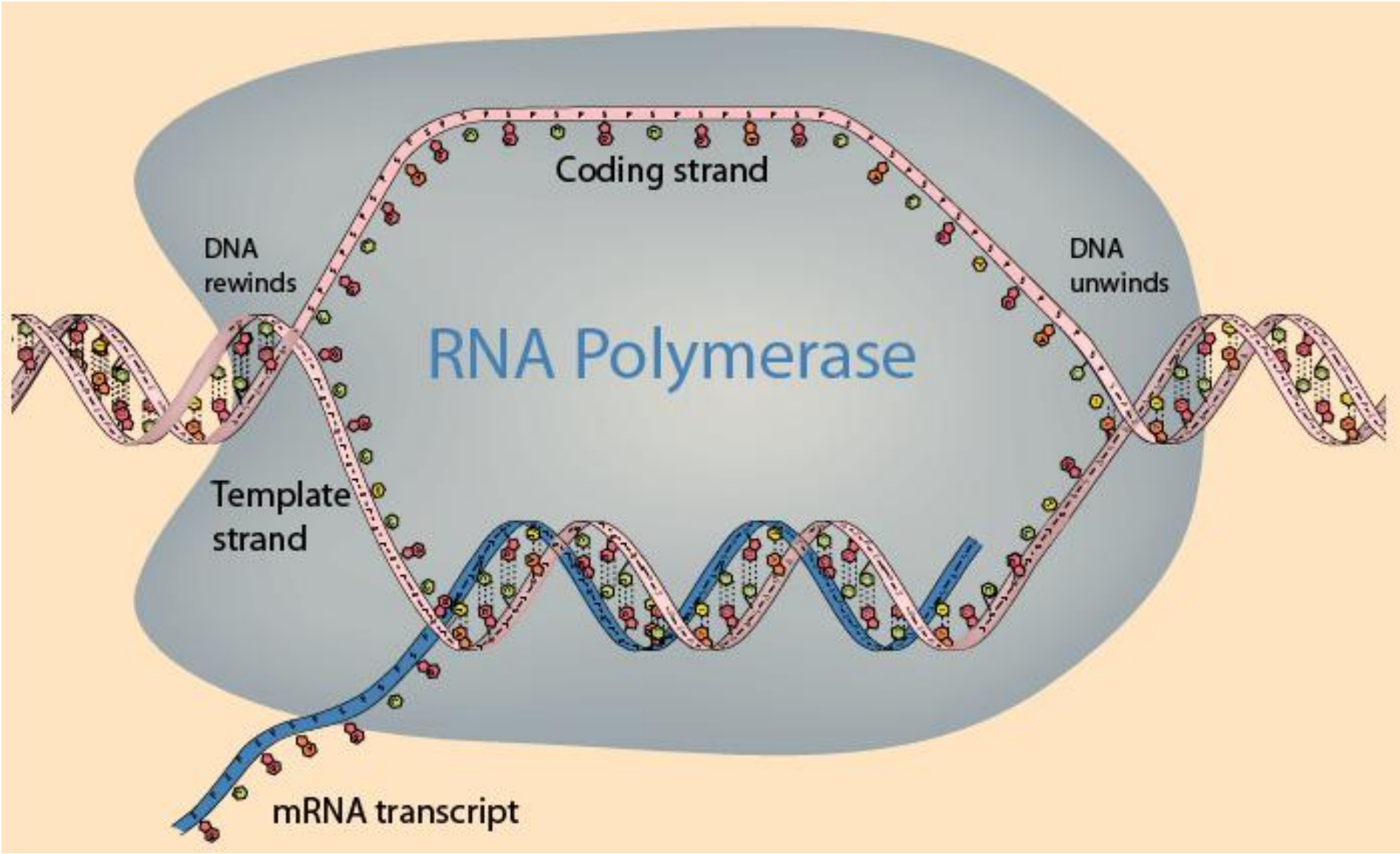
What Does DNA Do?

- Part of DNA codes for Proteins
  - ⋆ Gets copied to messenger RNA through transcription
  - ⋆ mRNA then gets translated to proteins
- A lot of DNA does **not** code for protein
  - ⋆ Can code for RNA molecules (ribozymes)
  - ⋆ Used for controlling the cell
  - ⋆ Can be junk—caused by replication
  - ⋆ We're not quite sure

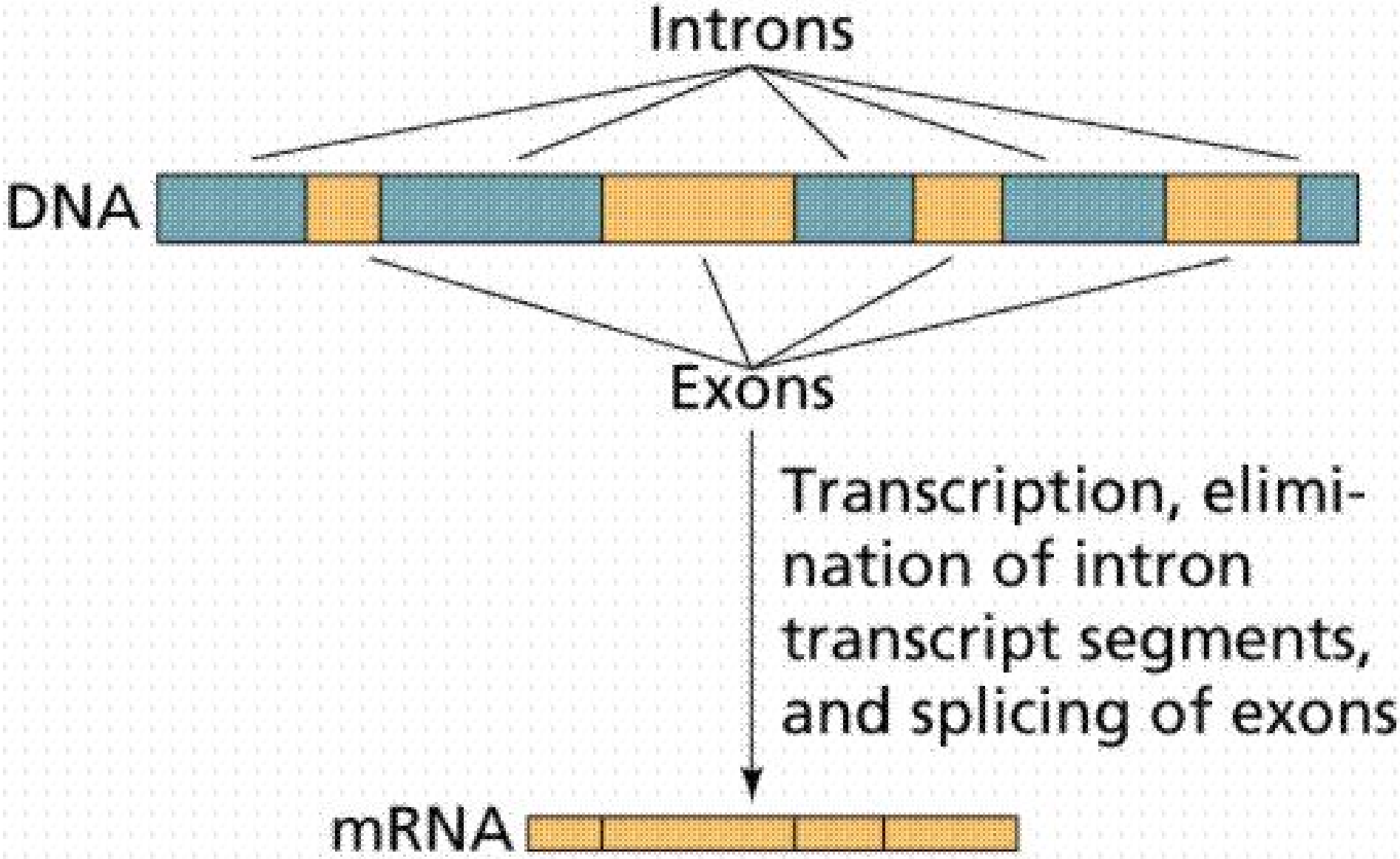
RNA and DNA


Cytosine
C
Guanine
G
Adenine
A
Uracil
U
Nucleobases of RNA
Nucleobases
Base pair
helix of sugar-phosphates
RNA
Ribonucleic acid
DNA
Deoxyribonucleic acid
Cytosine
C
Guanine
G
Adenine
A
Thymine
T
Nucleobases of DNA

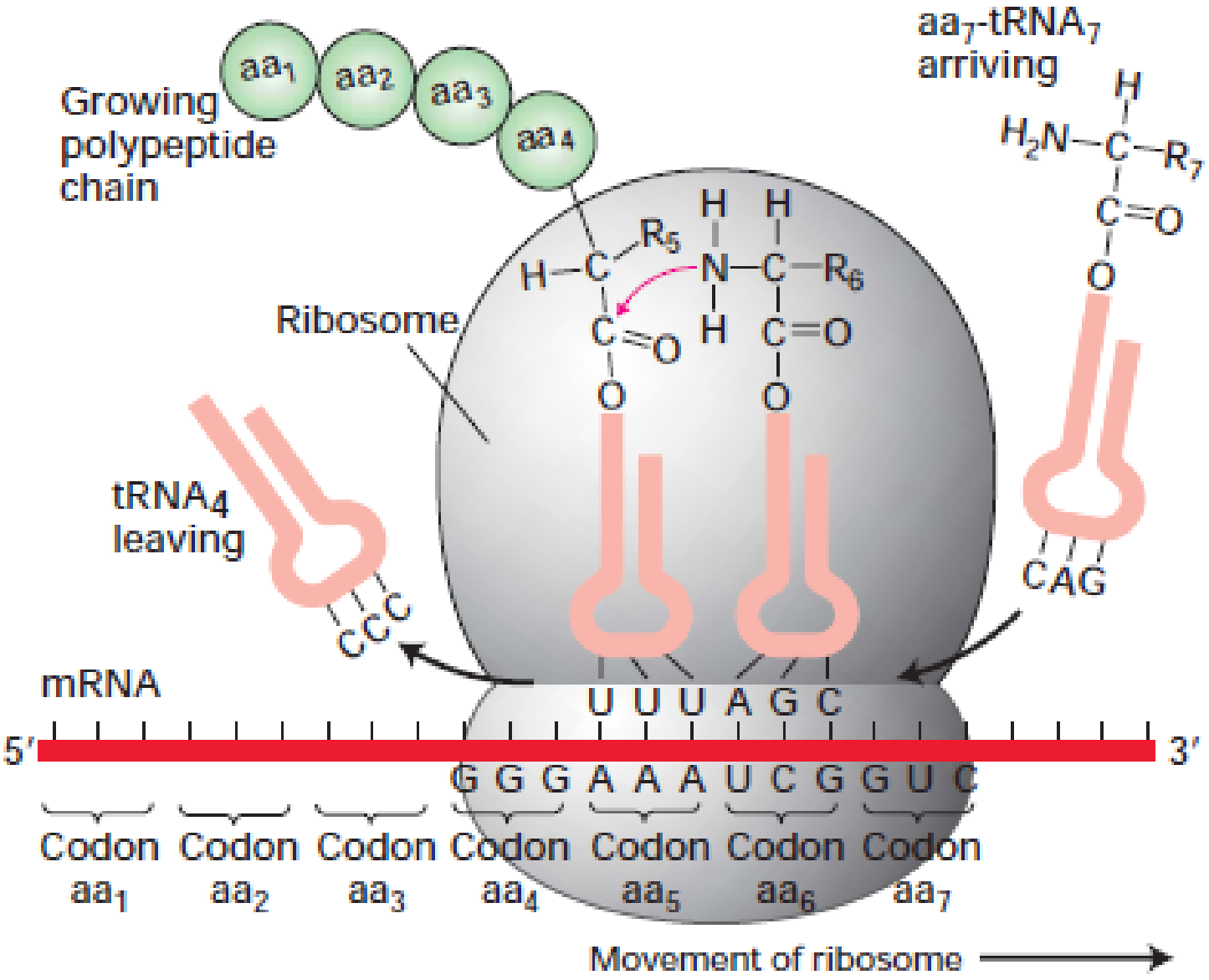
Transcription by RNA Polymerase in Action



Exons and Introns



mRNA to Protein Translation



RNA Triplet to Amino Acids

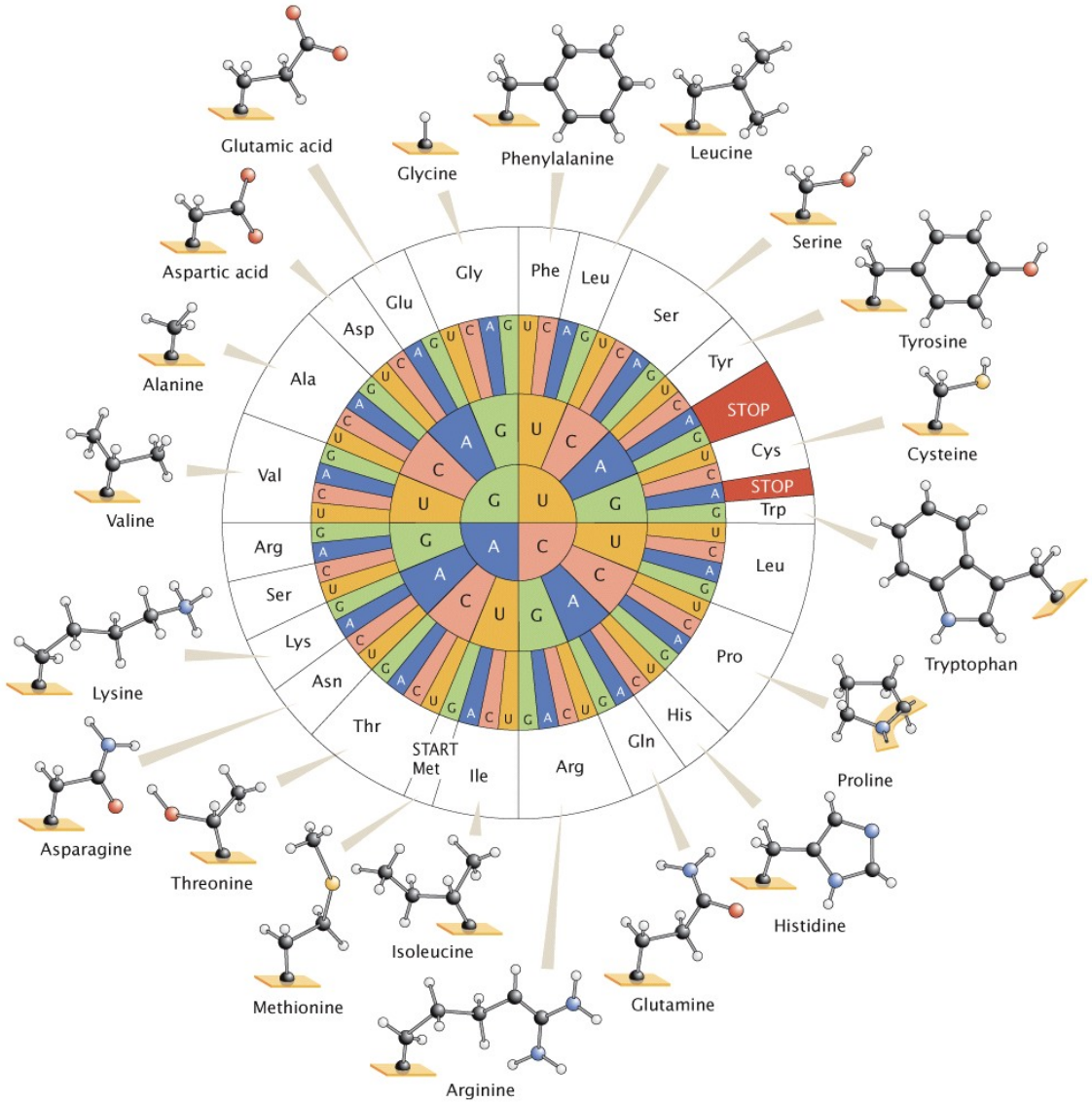
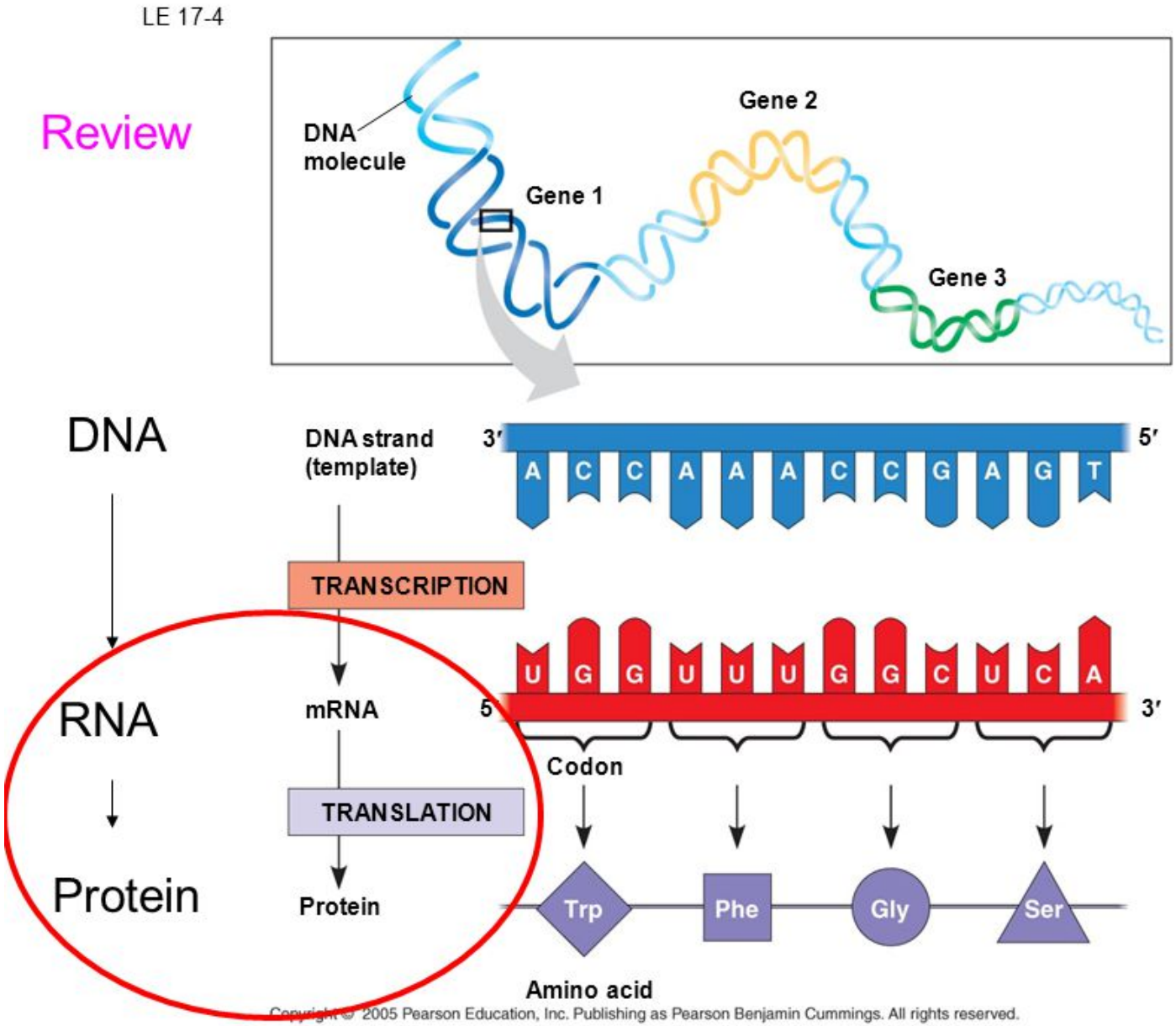


Figure 1.4 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

DNA to Protein



What Do Proteins Do?

- Proteins are the main building block of cells
- The can form the skeleton (e.g. lipid layer of membranes)
- They acts as chemical machines (enzymes)
- They can even act as mechanical machines
- They are a large part of the control mechanism
- Not everything in the cell is a protein
  - ⋆ E.g. there is DNA and RNA
  - ⋆ RNA can fold up like proteins and act as catalysts (ribosymes)

Computational Challenges

- A lot of computational challenges had to be addressed to understand the genome
- Need to find coding regions (exons)
- Need to reconstruct DNA sequence from sequencing data
- Need to compare different sequences
- We will cover these issues during the course (and more beside)

Conclusion

- We've just started to look at molecular biology
- We've covered inheritance through DNA
- How to produce proteins from DNA
- We've still got to look at how cells work